

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0480

Red Hill 345 kV
July 01, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0480. The Interconnection Customer proposes a 491.4 MW natural gas combined cycle facility interconnecting at the Red Hill 345 kV substation in the Virginia area. The request was received on May 26, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,575,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bear Island to Dunn Siding 345 kV	Leavitt to Birchwood 345 kV	140.1	1,885	7.4
Leavitt to Birchwood 345 kV	Bear Island to Dunn Siding 345 kV Line	136.2	2,046	10.1
Dunn Siding to Fredericksburg 345 kV Line	Nauvoo to Birchwood 345 kV Line	136.0	1,345	6.3
Adrian Energy Associates Llc 345/138 kV XFMR	Taber to Glenannan 345 kV	133.8	750	5.1
Riva Rd to Shingletown 345 kV	Taber to Glenannan Substation 345 kV	127.8	1,356	7.1
Bells Mill to Beverly Farms 138 kV line	Dranesville to Tiger Bay 345 kV	127.7	413	5.5
Nauvoo to Roberts Sw Sta 345 kV	Nauvoo to City West Diesel Plant 345 kV	126.7	1,290	8.3
Roberts Sw Sta to Glenannan 345 kV	Glenannan to Cal Sierra Development 345 kV line	126.5	1,223	9.0
Nauvo to City West Diesel Plant 345 kV Line	Nauvoo to Roberts Sw Sta 345 kV	117.4	1,338	7.3
Tombstone to Marcus Hook Energy Lp 500 kV	Sharyland to Tombstone 345 kV line	116.3	2,839	6.9
Chancellor Row to Bear Island 345 kV	Leavitt to Birchwood 345 kV	116.0	1,908	7.9
Glenannan to Cal Sierra Development 345 kV	Glenannan to Roberts Sw Sta 345 kV L	115.3	1,319	8.2
Wilkins to Highgrove 138 kV line	Dixleta Substation to Taber 345 kV Line	115.0	794	5.6
Wilkins to Wheelabrator Baltimore Refuse 138 kV	Taber to Dixleta 345 kV line	113.5	839	5.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rustic to Hughson 345 kV	Dixleta to Rustic 345 kV line	113.1	1,970	13.2
Dixleta 345-138 kV Transformer Bank T1	Dixleta to Taber 345 kV line	112.2	627	5.1
Dixleta 345/138kV Autotransformer T2	Taber to Dixleta 345 kV	112.2	627	5.1
Dixleta Substation 345/138 kV TX Bank 3	Taber to Dixleta 345 kV	112.2	627	5.1
Gordonsville Energy Lp to Littlefork 138 kV	Louisa Generation Facility to Fluvanna P.S 345 kV	111.9	355	6.4
Timken Mercy to Red Hill Substation 138 kV	Red Hill to Fluvanna P.S 345 kV	111.5	304	48.6
Lake Creek to Sierra Pacific Lumber Substation 138 kV	Glenannan to Taber 345 kV Line	111.4	997	5.1
Nauvoo Substation to Birchwood 345 kV line	Fredericksburg to Dunn Siding 345 kV Line	111.1	2,129	10.6
Carrabelle to Louisa Generation Facility 345 kV	Louisa Generation Facility to N Anna 345 kV	111.0	1,907	25.3
Bixby Hill Substation to Tombstone 500 kV	Sharyland to Tombstone 345 kV Line	110.3	3,937	8.8
Monroe e to Cedar Crk 345 kV	Co Line Rd to Monroe Ne 500 kV	109.6	1,144	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bear Island to Dunn Siding 345 kV	Rebuild line with higher-capacity conductor	\$55,580,000	2% (\$1,231,000)
Leavitt to Birchwood 345 kV	Rebuild line with higher-capacity conductor	\$96,258,000	3% (\$2,716,000)
Dunn Siding to Fredericksburg 345 kV Line	Rebuild line with higher-capacity conductor	\$69,173,000	3% (\$2,335,000)
Adrian Energy Associates Llc 345/138 kV XFMR	Replace transformer with higher-rated unit	\$11,997,000	20% (\$2,355,000)
Riva Rd to Shingletown 345 kV	Rebuild line with higher-capacity conductor	\$91,526,000	4% (\$3,443,000)
Bells Mill to Beverly Farms 138 kV line	Rebuild line with higher-capacity conductor	\$2,032,000	4% (\$89,000)
Nauvoo to Roberts Sw Sta 345 kV line	Rebuild line with higher-capacity conductor	\$36,807,000	5% (\$1,718,000)
Roberts Sw Sta to Glenannan 345 kV	Rebuild line with higher-capacity conductor	\$75,044,000	4% (\$2,709,000)
Nauvo to City West Diesel Plant 345 kV Line	Rebuild line with higher-capacity conductor	\$39,480,000	4% (\$1,737,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tombstone Substation to Marcus Hook Energy Lp 500 kV	Rebuild line with higher-capacity conductor	\$135,312,000	3% (\$3,883,000)
Chancellor Row to Bear Island 345 kV Line	Rebuild line with higher-capacity conductor	\$42,965,000	3% (\$1,213,000)
Glenannan to Cal Sierra Development 345 kV	Rebuild line with higher-capacity conductor	\$37,040,000	4% (\$1,362,000)
Wilkins to Highgrove 138 kV line	Rebuild line with higher-capacity conductor	\$552,000	3% (\$19,000)
Wilkins to Wheelabrator Baltimore Refuse 138 kV	Rebuild line with higher-capacity conductor	\$1,293,000	3% (\$44,000)
Rustic to Hughson 345 kV	Rebuild line with higher-capacity conductor	\$53,960,000	3% (\$1,553,000)
Dixleta 345-138 kV Transformer Bank T1	Replace transformer with higher-rated unit	\$15,684,000	4% (\$623,000)
Dixleta 345 kV/138 kV Autotransformer Unit 2	Replace transformer with higher-rated unit	\$23,515,000	4% (\$934,000)
Dixleta Substation 345/138 kV TX Bank 3	Replace transformer with higher-rated unit	\$20,392,000	4% (\$810,000)
Littlefork to Gordonsville Energy Lp 138 kV	Rebuild line with higher-capacity conductor	\$2,793,000	11% (\$304,000)
Timken Mercy to Red Hill Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,732,000	78% (\$1,347,000)
Sierra Pacific Lumber to Lake Crk 138 kV line	Rebuild line with higher-capacity conductor	\$2,018,000	4% (\$91,000)
Nauvoo to Birchwood 345 kV	Rebuild line with higher-capacity conductor	\$60,979,000	4% (\$2,410,000)
Carrabelle to Louisa Generation Facility 345 kV	Rebuild line with higher-capacity conductor	\$153,090,000	11% (\$16,774,000)
Tombstone to Bixby Hill 500 kV	Rebuild line with higher-capacity conductor	\$126,548,000	3% (\$3,640,000)
Monroe Ne to Cedar Crk 345 kV	Reconductor line	\$7,738,000	3% (\$235,000)
	Total allocated to project		\$53,575,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.